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Coupled hydrodynamic-structural analysis on reuseable launch vehicle landing on barge

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Sea-based landings offer a safer alternative to land, reducing risks to property and life while enabling the recovery of reusable launch systems to reduce launch costs. However, the dynamics of such landings pose significant challenges, particularly in minimizing structural damage and ensuring efficient recovery. This study examines the coupled hydrodynamic-structural response of a launch vehicle on a barge during touchdown, focusing on the coupled dynamics of the two systems. A time-domain model of the barge is developed using the Cummins equation, incorporating hydrodynamic parameters added mass and damping determined using HAMS. The launch vehicle is modelled as an equivalent beam and coupled to the barge, enabling the simulation of key transient behaviour during offshore landing events while allowing parametric studies and rapid design iterations. The sensitivity of the transient response of the coupled model is analysed with variability in the touchdown velocity and landing location. The simulation framework developed captured key hydrodynamic-structural excitations of the system. Results indicate that offset landing induces coupling between heave and pitch modes, while structurally exciting the bending modes. However due to maintain-ing of sufficient bandgap between the hydrodynamic and structural modes the coupling effect is reduced and response is of low magnitude.

1. INTRODUCTION

The space industry, driven by the need to reduce costs and promote sustainable access to space, has undergone a shift towards using reusable launch vehicles (RLVs). Private aerospace companies like SpaceX and Blue Origin have been at the forefront of this shift. For instance, SpaceX reports that reusing Falcon 9 boosters can reduce launch costs by up to 80% [1]. The vertical landing of spacecraft has been extensively studied in the context of space exploration, with many papers focusing on novel landing gear designs for planetary surfaces [2–4]. Within this context, offshore landings have emerged as a critical strategy for optimizing mission performance, reducing fuel consumption, mitigating trajectory reversal complexities, and improving operational safety, making it a viable long-term solution for reusable launch systems [5]. However, the literature on offshore landing is scarce. Although some studies have proposed models for floating platform landings [1], detailed dynamic analysis is notably absent. The dynamic motion of floating platforms under wave, wind, and current loads further complicates precision landing and post-touchdown stability and must be analyzed to ensure safe recovery of the RLV. Modeling the soft landing of launch vehicles onto floating platforms requires a comprehensive understanding of the coupled interaction between the vehicle's structural flexibility and the platform's hydrodynamic response to waves. Literature in offshore and aerospace engineering provides several approaches for capturing these complex interactions, each balancing fidelity, computational effort, and practical applicability.

To ground our analysis in a real-world platform, we focus on the offshore landings of the Falcon 9. The Falcon 9 is a two-stage RLV developed by SpaceX. Due to the wide media coverage of its missions, this RLV has been selected as a case study to identify the possible modes of failure during an offshore landing. For offshore recovery missions, SpaceX utilized converted deck barges, notably the MARMAC 302, for the Falcon 9. Originally a flat-top cargo barge, the MARMAC 302 was adapted into an Autonomous Spaceport Drone Ship (ASDS) to facilitate RLV landings far offshore. The specifications of both are summarized in Table 1.

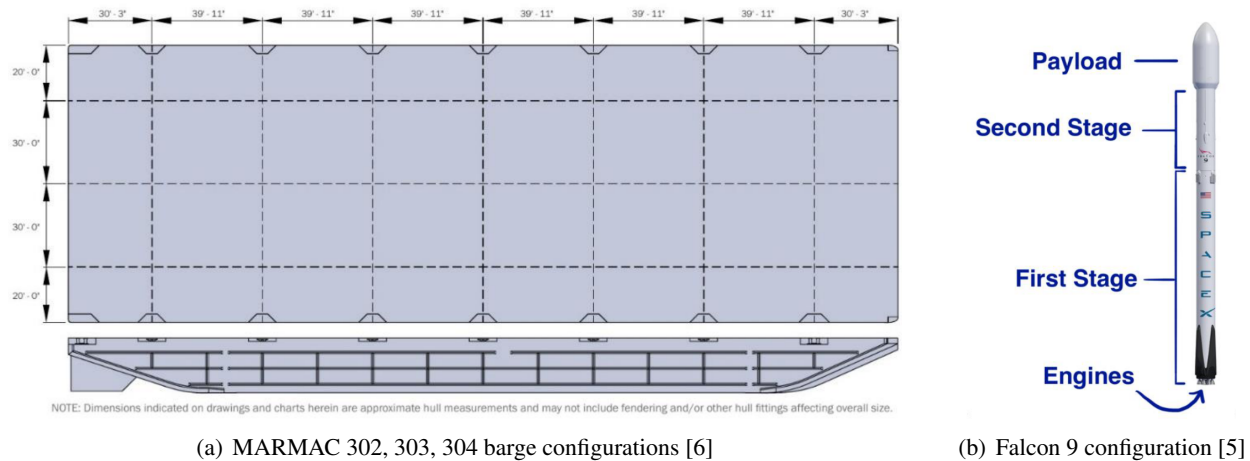


Figure 1: Model configurations used for the landing simulation

Early attempts at Falcon 9 barge landings highlighted the complex interplay between the platform's wave-induced motions and the vehicle's structural dynamics during touchdown. Failure modes observed include:

- Hard landings: Uncontrolled descent due to loss of thrust control leading to structural failure.
- Offset touchdowns: Lateral position errors during landing lead to large impact moments and uneven

load distribution on the landing legs, which may lead to structural failure. An example of this is shown in Fig. 2.

- Post-landing toppling: Platform motions of roll and pitch in high sea states may lead to high inclination angles. These, along with the relative motion of the launch vehicle, can lead to the toppling of the launch vehicle, resulting in the failure of recovery as shown in Fig. 3.



Figure 2: Sequence showing Falcon 9 landing offset to the designated landing site [1].



Figure 3: Sequence showing Falcon 9 toppling after successful touchdown on a moving barge [1].

This complex interaction between barge hydrodynamics, atmospheric descent dynamics, and vehicle flexibility makes offshore landing one of the most challenging phases of Falcon 9 recovery. Accurate modeling of these coupled dynamics would provide critical insights for designing robust recovery strategies and improving success rates. The final phase of rocket recovery—landing—is critical to the success of reusable launch systems. Achieving a soft landing minimizes structural damage and ensures the viability of subsequent missions. Offshore landings, in particular, involve complex interactions between the rocket and the dynamically moving platform, further complicated by hydrodynamic effects such as added mass and damping. While land-based recoveries have been extensively studied, offshore landings remain less explored. A clear research gap exists in understanding the coupled dynamics during touchdown on floating structures. Addressing this gap is essential for optimizing sea-based recovery strategies and improving the overall reliability of reusable launch operations. The objectives and scope of this study are to develop a theoretical framework for modeling the hydrodynamic and structural behavior of the landing of the RLV on a barge. The insights gained are expected to inform design improvements and operational guidelines for offshore recovery, contributing to advancing reliable and efficient reusable spaceflight technologies.

2. MODEL DEVELOPMENT

In launch vehicle landing dynamics, fully coupled models offer unmatched fidelity but are often computationally prohibitive, particularly for long simulations or large-scale parametric studies. Simplified structural representations — such as equivalent beam models — have been increasingly adopted as a practical alternative. These approaches idealize the vehicle as a cantilever beam with distributed mass and stiffness properties, capturing primary structural modes critical during touchdown. A similar modeling strategy has been employed for the analysis of vibration transmission in marine systems, where a flexible cutter shaft

assembly on a floating barge was modeled using finite element techniques to study dynamic responses under wave-induced and impulsive loads [7, 8]. Although applied to a different operational context, the study demonstrated the importance of capturing structural flexibility, hydrodynamic interaction, and coupling effects to accurately predict transient dynamic behavior. This study considers two system configurations: a simplified box barge and a trapezoidal approximation of the MARMAC 302 platform, modeled after SpaceX's Just Read the Instructions (JRTI) autonomous spaceport drone ship (ASDS). Two geometries are analyzed whose geometric properties are listed in Table 1.

Table 1: Barge Geometry and Mass Parameters

Parameter	Box Barge	MARMAC 302	Unit
Barge Mass	260800	7129486	kg
Barge Length	44	91.44	m
Barge Breadth	4	30.48	m
Barge Depth	4	6.0198	m
Design Draft	2	2.9718	m
RLV Mass	100000	20000	kg
RLV Height	22	40.9	m
RLV Radius	1.6	1.83	m
RLV Thickness	4	5	mm

In both the MARMAC302 and the box barge, the masses are chosen such that the hull is nearly half immersed, which can be seen by the ratio between the depth and the design draft for the barges. To explore a worst-case scenario, the RLV and barge mass are redistributed to produce a heavy top loading in the box barge case.

3. THEORETICAL MODELLING

This section presents the coupled time-domain simulation framework to model the touchdown dynamics of an RLV on a floating recovery platform. The workflow consists of four stages: barge geometry modeling and meshing, hydrodynamic coefficient computation, structural modeling and coupling of the RLV, and numerical integration of the coupled dynamic equations.

A. HYDRODYNAMIC MODELLING

The dynamic response of floating platforms to wave excitation is traditionally analyzed using linear frequency-domain methods based on potential flow theory using hydrodynamic solvers like WAMIT. A more flexible approach involves using linear time-domain models based on the Cummins equation, which captures fluid memory effects through convolution integrals. Ogilvie (1964) established the relationship between the frequency-domain added mass and damping coefficients and the time-domain retardation functions. These relations allow conversion between time- and frequency-domain hydrodynamic parameters, enabling flexibility in simulation techniques. This hybrid frequency-time-domain framework enables modeling nonlinear forces as external contributions, thereby enhancing simulation fidelity for floating platform dynamics [9].

In the current study, the hydrodynamic coefficients are computed using Hydrodynamic Analysis of Marine Structures (HAMS), an open-source solver developed for the analysis of wave diffraction and radiation of three-dimensional floating structures. [10, 11]. HAMS solves the linear potential flow problem based on

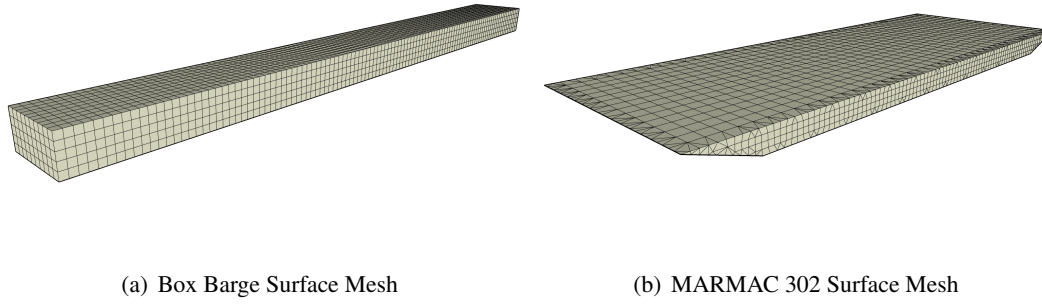


Figure 4: Hydrodynamic surface mesh generated using Gmsh

boundary integral equations, assuming inviscid, incompressible, and irrotational flow under small-amplitude wave conditions to obtain the frequency domain added mass (A_{ij}) and radiation damping (B_{ij}). Additional parameters such as wave excitation forces, response amplitude operators (RAOs), and hydrodynamic pressure distributions are also computed, though they are not used in the present study. The motion of the floating barge is modeled using the Cummins Equation [9]. Port-starboard symmetry is assumed along the longitudinal (Y) axis. The complete geometry is modeled for the computation of hydrodynamic parameters in all six degrees of freedom (DOFs). Based on geometric symmetry and loading, the critical dynamics of the system are observed in the vertical XZ plane, and coupling to modes in the horizontal plane is assumed negligible. Therefore, the barge motion is modeled using a reduced 3 DOF system (surge, heave, and pitch). The reduced governing equation for the motion is given by,

$$\sum_{j=1,3,5} \left[(M_{ij} + A_{ij}) \ddot{x}_j + \int_0^t K_{ij}(t - \tau) \dot{x}_j(\tau) d\tau + C_{ij} x_j \right] = F_i^{\text{ex}} \quad (i = 1, 3, 5) \quad (1)$$

where the indices i and j correspond to the degrees of freedom 1, 3, 5, representing surge, heave, and pitch, respectively, considered in the analysis. For each mode i , the summation captures the dynamic coupling with other relevant modes j . Here, M_{ij} denotes the elements of the rigid body mass matrix, A_{ij} represents the added mass coefficients, and C_{ij} is the hydrodynamic restoring stiffness matrix. The displacement in mode j is given by $x_j(t)$, and $F_i^{\text{ex}}(t)$ is the wave excitation force acting in mode i . The convolution kernel $K_{ij}(t)$ accounts for radiation effects and is derived from the frequency-dependent radiation damping coefficients using the Ogilvie relation given by:

$$K_{ij}(t) = \frac{2}{\pi} \int_0^\infty B_{ij}(\omega) \cos(\omega t) d\omega \quad (2)$$

where B_{ij} is the frequency domain radiation damping. The drift in the surge direction is reduced by including a linear mooring stiffness. Surface meshes are generated using Gmsh with quality controls to ensure solver stability, as shown in Fig. 4.

i. Benchmarking Hydrodynamic Response Modeling

The time-domain model response is compared against a frequency-domain solution using the Fourier Transform Method. In this approach, the system response vector $\zeta(t)$ is reconstructed from frequency-domain transfer functions by applying inverse Fourier transforms, as given by:

$$\zeta(t) = \frac{2}{\pi} \int_0^\infty \text{Re} \left\{ \left(-s^2 \mathbf{M} + \mathbf{C} - s^2 \mathbf{A}(s) - is \mathbf{B}(s) \right)^{-1} \left(-\mathbf{M} - \frac{i}{s} \mathbf{B}(s) \right) is \zeta(0) \right\} \cos(st) ds \quad (3)$$

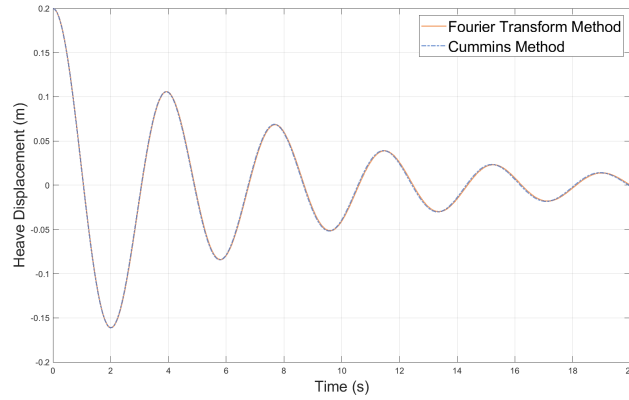


Figure 5: Motion Results Comparison between Cummins Equation and Fourier Transform Method

where $\mathbf{M}$ is the mass matrix, $\mathbf{C}$ is the hydrodynamic stiffness matrix, $\mathbf{A}(s)$ and $\mathbf{B}(s)$ are the frequency-dependent added mass and damping matrices and s is the frequency. This formulation provides a benchmark for assessing the accuracy of the convolution-based Cummins equation. The comparative results show good agreement, as shown in Fig. 5, affirming the fidelity of the hydrodynamic model for simulating transient platform responses under specified initial conditions, as also discussed in [12].

B. REDUCED STRUCTURAL MODEL

Equivalent beam-based models, when coupled with time-domain hydrodynamic simulations based on the Cummins equation, offer a favorable balance between simulation fidelity and computational efficiency. They enable the simulation of key transient behaviors during offshore landing events while allowing parametric studies and rapid design iterations, making them well-suited for the present investigation into reusable launch vehicle recovery dynamics.

The RLV is modeled as an equivalent cantilever beam with a cylindrical cross-section, assuming the material to be steel with a Young's Modulus of 210 GPa and homogeneous material properties. The dynamic stiffness values associated with the axial and bending modes are determined using the equivalent model. The bending is considered as a two-DOF model with transverse displacement and rotation, and the equivalent stiffness in the bending direction is given:

$$\begin{aligned} K_{11}^{eq} &= \frac{3EI}{L^3} \\ K_{55}^{eq} &= \frac{2EI}{L^2} \end{aligned} \quad (4)$$

where E is the Young's modulus, I is the area moment of inertia and L is the beam length. Similarly, the equivalent axial stiffness is given by:

$$K_{33}^{eq} = \frac{EA}{L}, \quad (5)$$

where A is the cross-sectional area. The RLV geometry from Table 1 is used to calculate the equivalent stiffness values using Eq. 4 and 5.

The RLV structural natural frequencies are assumed to be 10 Hz in bending and 30 Hz in axial directions and are selected to be far apart from the natural frequencies of the barge. The heave natural frequency was assumed to account for the difference between axial and bending modes. The stiffness values and natural frequencies are used to compute the equivalent mass M^{eq} of the launch vehicle in each mode. Due to the

changed mass representation in the heave mode, the static draft will change. This effect is then balanced by adding a mass correction to the barge in the heave mode. The equivalent mass and mass corrections for all modes are summarized in Table 2.

Table 2: Dynamic Properties of the Launch Vehicle Across Platform Cases

Barge Case	Mode	ω_j^{nat} (Hz)	K_j (N/m)	M_{jj}^{eq} (kg)	Corrected Mass (kg)
Box Barge	Transverse	10	7.85×10^5	48.03	—
Box Barge	Axial	30	3.84×10^8	10803.24	3.50×10^5
Box Barge	Rotational	10	6.12×10^7	704.47	—
MARMAC 302	Transverse	10	5.52×10^4	13.98	—
MARMAC 302	Axial	30	2.95×10^8	8307.97	7.14×10^6
MARMAC 302	Rotational	10	1.50×10^6	381.08	—

C. COUPLED STRUCTURAL HYDRODYNAMIC MODEL

The coupled dynamics of the RLV and the barge are governed by the coupled second-order differential equations of motion. The barge and the RLV were coupled using springs of equivalent stiffness values along the degrees of freedom as shown in Fig. 6.

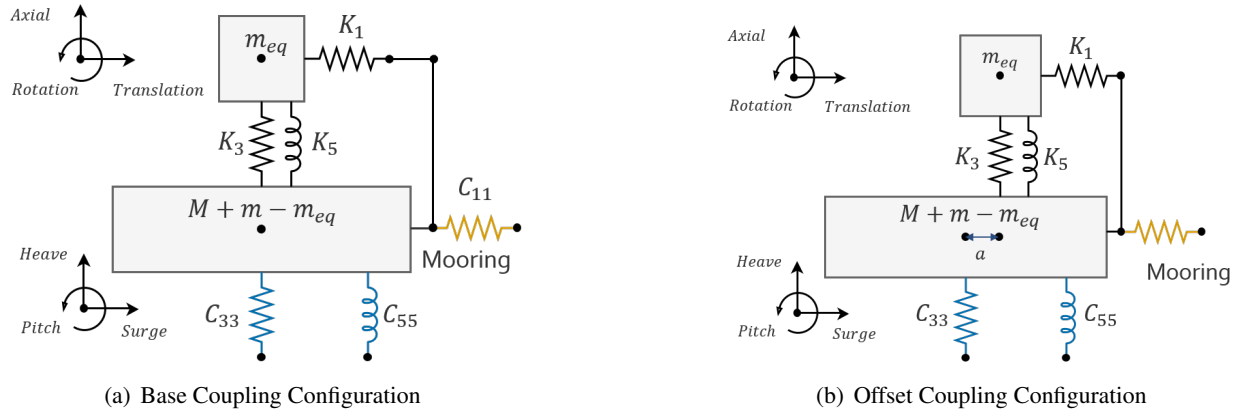


Figure 6: Touchdown interface coupling between the launch vehicle and the recovery platform

The system matrices are partitioned to represent the barge and vehicle subsystems separately. The convolution term is only associated with the barge and is hence computed separately and then added to the equation, keeping the corresponding term for the vehicle damping as zero. The final coupled equation of motion is given as:

$$\begin{bmatrix} \mathbf{M}_b^{\text{corr}} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_v^{eq} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{x}}_b \\ \ddot{\mathbf{x}}_v \end{bmatrix} + \begin{bmatrix} \mathbf{K}_b + \mathbf{K}_c & -\mathbf{K}_c \\ -\mathbf{K}_c^T & \mathbf{K}_v + \mathbf{K}_c \end{bmatrix} \begin{bmatrix} \mathbf{x}_b \\ \mathbf{x}_v \end{bmatrix} + \begin{bmatrix} \mathbf{F}_{\text{conv}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_b \\ \mathbf{F}_v \end{bmatrix}, \quad (6)$$

where:

- $\mathbf{x}_b, \mathbf{x}_v$: Displacement vectors for the barge and vehicle DOFs,
- $\mathbf{M}_b^{\text{corr}}$: Corrected barge mass matrix (including the launch vehicle's contribution to heave mass),
- $\mathbf{M}_v^{eq}$: Diagonal vehicle modal mass matrix,

- $\mathbf{K}_b, \mathbf{K}_v$: Stiffness matrices for the barge and vehicle,
- $\mathbf{K}_c$: Coupling stiffness matrix derived from equivalent beam properties,
- $\mathbf{F}_{\text{conv}}$: Hydrodynamic convolution force vector accounting for memory effects,
- $\mathbf{F}_b, \mathbf{F}_v$: External force vectors acting on the barge and the launch vehicle, assumed to be zero.

The fully coupled system is implemented in MATLAB and solved using the `ode23t` solver, appropriate for stiff ODE systems. Simulations are performed across various touchdown scenarios, including both base-centered and offset landing conditions, under different impact velocities.

4. SENSITIVITY OF LANDING LOCATION

Two touchdown conditions are modeled:

- Center landing: RLV lands vertically above the barge center of mass (CoM).
- Offset landing: RLV lands with lateral displacement, inducing trim and altering the CoM of the system and center of buoyancy of the barge. The offset distance a is the perpendicular distance between the midplane of the barge and the RLV's axis in the static condition. The resulting configuration with a static trim is shown in Fig. 7(b).

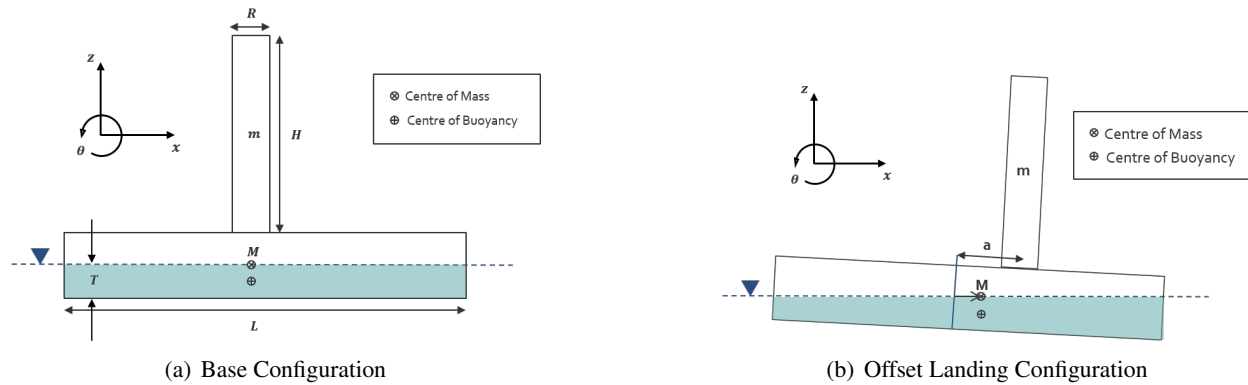


Figure 7: Schematic representations of barge touchdown configurations

The change in the CoM of the system and the resulting change in the static geometry are computed for both cases using MATLAB. The hydrodynamic parameters of these changed static conditions are used in the analysis of the offset cases. X and Z coordinates of the changed CoM are listed in Table 3. The offset geometries with their changed CoM and center of buoyancy are shown in Fig. 8.

Table 3: HAMS Input Parameters for Platform Cases

Parameter	Box Barge (5m Offset)	MARMAC 302 (30m Offset)	Unit
CoM X	1.3858	0.0839	m
CoM Z	-0.0238	-0.0025	m

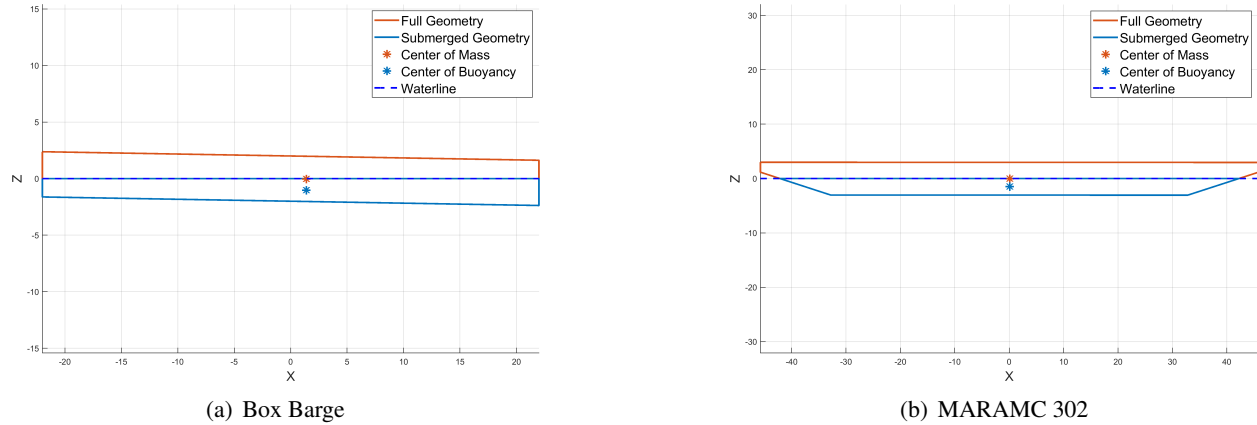


Figure 8: Offset geometry plots

5. RESULTS AND DISCUSSIONS

This section presents the results of the coupled dynamics simulations of the launch vehicle (RLV) and barge during soft landing under four configurations:

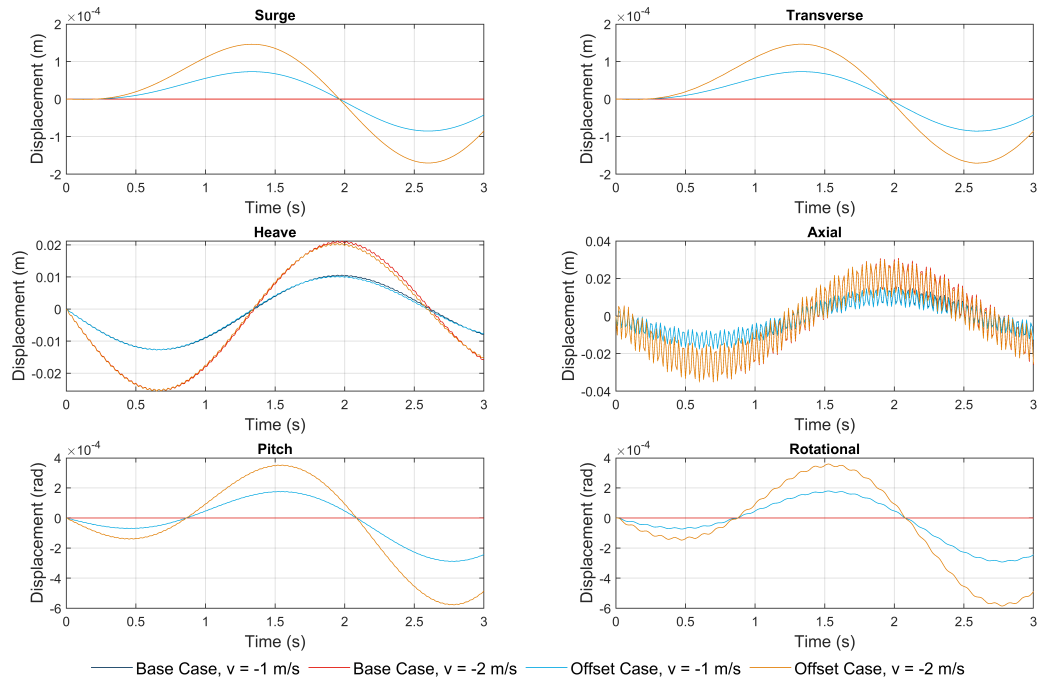
- Box Barge: Center landing
- Box Barge: Offset landing ($a = 5$ m)
- MARMAC302: Center landing, and
- MARMAC302: Offset landing ($a = 30$ m).

Figure 9 shows the time and frequency response of displacement for the Box Barge with landing at the center. The sensitivity to initial velocity is analyzed, which influences the bandwidth of excitation. The low-frequency peak observed in the response is due to the hydrodynamic characteristics of the vessel. The higher frequency is due to the axial and bending modes, and pitching is not excited due to the central landing. Figure 10 shows the MARMAC 302 response. Due to the variation in the hull form, the hydrodynamic characteristics are altered accordingly. Since the structural and hydrodynamic frequencies are widely spaced, the responses do not mutually add up to produce a high response.

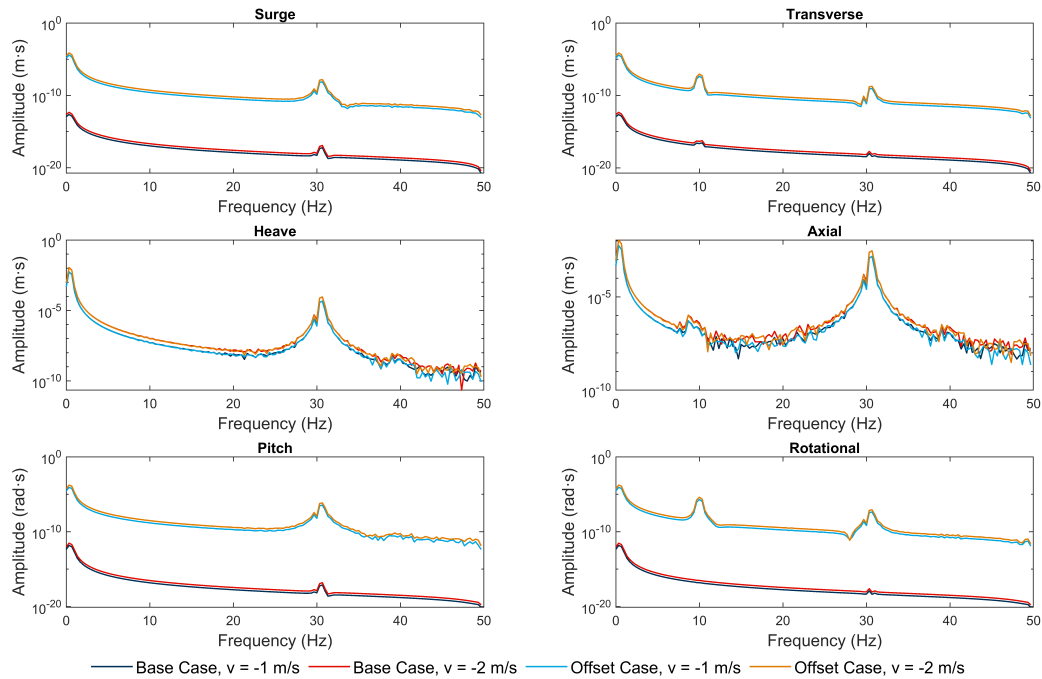
Figures 9 and 10 show that the pitch response of the hull and transverse motion of the RLV are higher for offset landing in comparison to the central landing. The structural response at 10 Hz is more prominent. This indicates that the offset landing causes coupling of heave and pitch hydrodynamic responses. Some of the energy from the pitch response is transmitted to the bending of the RLV. Similar to the case in central landing, since the hydrodynamic and structural modes have a wide band gap, the structural response in bending and axial direction of the RLV is less significant compared to the hydrodynamic response of heave and pitch.

6. CONCLUSION

This work has presented a coupled time–domain analysis of the soft landing of an RLV on two floating recovery platforms (Box Barge and MARMAC302) under both central and offset touchdown scenarios. The sensitivity of the landing location is analyzed by determining the structural and hydrodynamic response. The results from the study indicate that the offset landing can induce a coupling of heave and pitch response hydrodynamically. Structurally, the offset landing can excite the bending modes. However, the structural

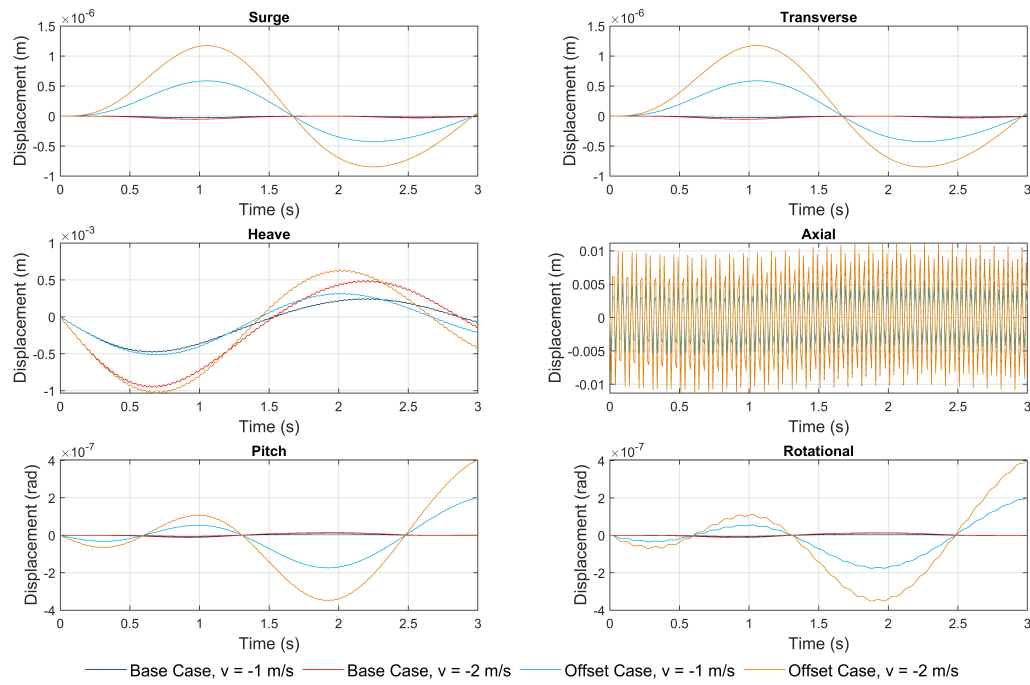


(a) Displacement response corresponding to the DOF in the vertical plane.

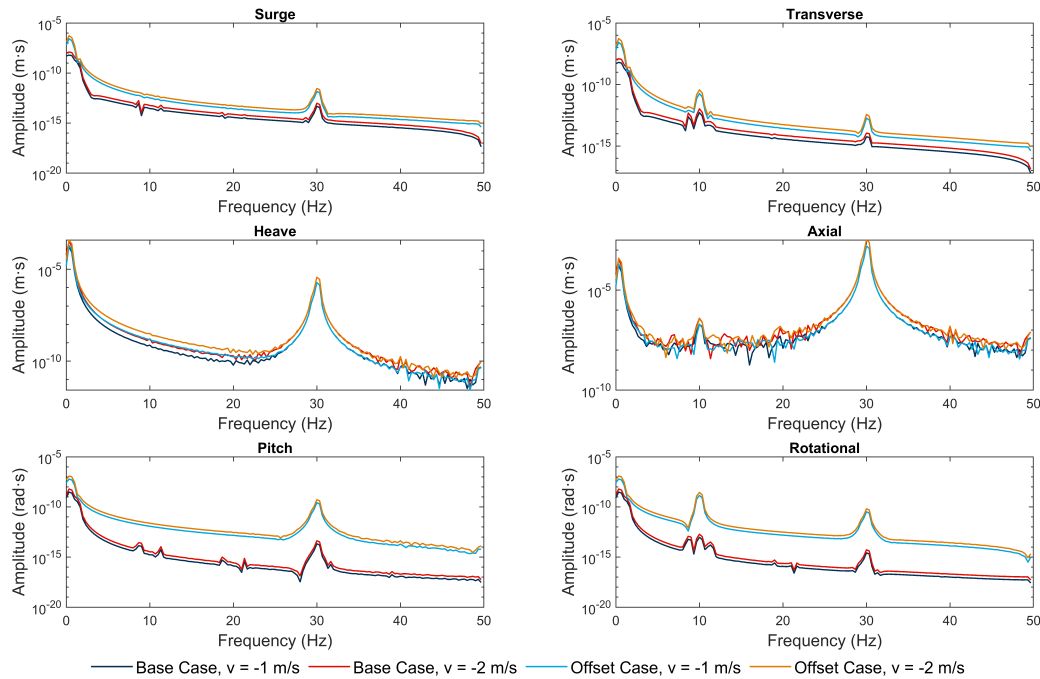


(b) Frequency response of Displacement corresponding to the DOF in the vertical plane.

Figure 9: Box Barge - Response comparison corresponding to mid and offset landing. Surge, Heave and Pitch modes corresponds to the Box barge. Transverse, Axial and Rotational corresponds to the flexural modes of the RLv.



(a) Displacement response corresponding to the DOF in the vertical plane.



(b) Frequency response of Displacement corresponding to the DOF in the vertical plane.

Figure 10: MARMAC 302 - Response comparison corresponding to mid and offset landing. Surge, Heave and Pitch modes corresponds to the MARMAC 302. Transverse, Axial and Rotational corresponds to the flexural modes of the RLV.

and hydrodynamic response coupling can be reduced by maintaining a sufficient bandgap between the hydrodynamic and the structural modes of the system. Future investigation will focus on developing a detailed mechanism for a soft landing of the RLV on a barge. This will primarily involve modeling the RLV's soft landing using compliant nonlinear mechanisms, which could reduce the response even in the case of a reduced bandgap. For further refinement of the model, wind loading and random wave loading can be considered in the response analysis.

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